

Geographies of Experiment/Experimental Geographies: A Rough Guide

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Abstract

Geographers have increasingly employed the notion of experiment to describe the explorative styles of researching and thinking that they have elaborated in response to the complexities that characterise the present world. This article discusses the shifting uses of the notion in recent contributions, tracing out the ethical, methodological and theoretical sensibilities that it brings into geography. The article falls into two parts, the first reviewing *geographies of experiment*—that is, accounts exploring diverse empirical sites of experimentation, from scientific laboratories to performance art and urban spaces. The second part concentrates on *experimental geographies* which involve attempts by human geographers and other social and cultural researchers to reconsider their practices as experimental. Although the notion of experiment might take on varied functions and meanings, the article indicates that it offers plenty of inspiration for geographical learning and more collaborative and responsive modes of researching and thinking.

1. Introduction: A Society of Experimentation¹

All around the world geographers are now both studying and taking part in [...] spatial experiments [...] These experiments range far and wide; all the way from the [...] kinds of experiments that are meant to perform everyday life differently to the kinds of experiments which are trying to do new kinds of global (Thrift 2009, 100).

What might the notion of experiment have to offer geographers? It has recently appeared in diverse conceptual guises across the social sciences, including ‘experimental society’ (Davies 2010, 668), ‘collective experimentation’ (Latour 2004a, 196; original emphasis), ‘experimental disposition’ (Ronell 2005, 205), ‘self-experimental society’ (Gross and Krohn 2005, 65), ‘experimental turn’ (Powell and Vasudevan 2007, 1791), ‘ethos of experimentation’ (Latham and Conradson 2003, 1904) and ‘cosmopolitical experiments’ (Hinchliffe et al. 2005, 644). Each of these concepts evokes slightly different concerns, but together they point to a broader shift in the way experimentation is understood both as an empirical phenomenon and an orientation to knowledge.

Among the above authors, there seems to be a shared sense that ‘experiments are no longer conducted in the laboratory but have become collective experiments that concern each and every one of us’ (Gross 2009, 93). Experimentation has emerged as a way for societies to act in an increasingly complex environment, where the challenges they encounter overflow institutional, regional and ontological boundaries. Dealing with issues such as climate change and sustainable development is viewed as requiring ‘new kinds of responsibility and responsiveness from scientists, politicians, artists and social scientists, as well as citizens’ (Davies 2010, 668), who now need to engage in a collaborative process of ‘learning by experimenting’ (Gross and Krohn 2005, 77).

Reading through these accounts, it becomes evident that the notion of experiment does not fall into neat disciplinary compartments but assumes the role of a ‘travelling concept’

(Bal 2002, 22–55) that introduces a range of orientations and sensibilities when put to work in different areas of enquiry. Partly because it may have so diverse functions and meanings, Powell and Vasudevan (2007, 1791) warn us against casual uses of the notion, as this may easily lead to little more than ‘slack metaphors’—for instance, we might justifiably want to ask whether experimental practices in scientific laboratories (Barad 2007), ecological restoration (Gross 2010), urban design (Evans and Karvonen 2011), performance art (Ingram 2012), global military industry (Graham 2011) and social work (Gross 2009) can possibly have anything in common.

The aim of this article is therefore to unfold some of the implications of the notion of experiment for geographers, *both* as an empirical phenomenon *and* as an orientation to knowledge, while at the same time contributing to recent reassessments of experimentation across the social sciences (see Davies 2010; Gross 2010; Guggenheim 2012; Last 2012; Powell and Vasudevan 2007; Rabinow 2012).² As will become clear, the notion of experiment does not so much trace out a clear-cut area of enquiry as a general ‘ethos’ (Latham and Conradson 2003, 1904) of researching and thinking that potentially fosters more collaborative and responsive modes of working as well as highlights possible ways of enhancing geographical learning. Bringing together contributions on the topic in geography and related fields, the article acts as a ‘rough guide’ (Bal 2002) into this emerging ethos of experimentation. ‘Rough’, because it can only gesture towards the vast terrain opened by experimentation. ‘Guide’, as it ‘can be used in any way you like, following or wilfully ignoring any of the paths it maps’ (Bal 2002, 3).

There is an abundance of accounts of experimentation, but this brief overview limits itself to work that makes explicit use of the notion of experiment and offers reflective insight into the specific empirical and ethical, methodological and theoretical possibilities that it suggests for geographers. After Powell and Vasudevan (2007), the article is divided into two parts—the first, *geographies of experiment*, reviews research on experimental sites, from laboratories to arts and cities, whereas the second part, *experimental geographies*, reviews current attempts by geographers to rethink their concepts, ethics and practices as experimental, thereby envisioning new modes of doing geography. Although Powell and Vasudevan (2007, 1791) argue that ‘Experimental geographies [...] are not necessarily geographies of experiment’, this article suggests that these two areas may usefully complement one another—for example, exploring the forms of experimentation that presently engage natural scientists, artists and various others may offer geographers an opportunity to *learn* how to perform more productive experiments. As Latour (2004a, 196) writes, a ‘bad experiment is not one that fails, but one from which the researcher has drawn no lesson that will help prepare the next experiment’.

2. *Geographies of Experiment*

This part of the article, then, explores the varied *geographies of experiment* by directing ‘attention to the full range of bodies, texts and practices that constitute spaces of experimentation’ (Powell and Vasudevan 2007, 1790). Before visiting such experimental settings, it is essential to outline the ‘experimental turn’ (Ronell 2005, 9) discussed in recent work by various social scientists, who argue that experimental sites have proliferated in present societies, and that more people have become implicated in diverse forms of experimentation (see Callon et al. 2009; Davies 2010; Gross 2010; Hinchliffe et al. 2005; Latour 2004a):

a society of experimentation [...] builds its existence on certain kinds of experiments that are practiced outside the special domain of science. [This] is a deliberate intervention that is undertaken by a rapidly developing society that sets up institutional conditions of action without being able to

control their dynamics [...] As people try out things, they better understand what they are doing as well as what they plan for the future (Gross 2010, 173).

Researchers and theorists have increasingly pointed to the uncertainties generated by current configurations of culture and nature, science and technology, from concerns over climate change to nuclear waste (e.g. Callon et al. 2009; Latour 2004a). Such challenges often evade expert control and blur the institutional boundaries between politics, science and technology, thereby calling for more adaptive and collaborative solutions. Diverse 'experimental practices' (Gross 2010, 173)—including trials of sustainable technology in homes (Marres 2009), ecological restoration (Gross 2010) and collective flood-risk management (Whatmore and Landström 2011)—have therefore been elaborated by both experts and laymen as a flexible strategy for dealing with present uncertainties, while also engaging in the shaping of better futures. These experiments not only seek to cope with the unpredictabilities of the world, but encourage communities to learn from such 'unknowns' and to 'live and blossom with them' (Gross 2010, 34).

At the same time, it is essential to develop a 'geographical sensitivity to the spaces of actual experiments' (Powell and Vasudevan 2007, 1790), as we cannot assume that 'the experiment is everywhere and always the same' (Davies 2010, 668). And much like any other practice, experiments can be put to ethically questionable uses: detention facilities and concentration camps (Bauman 1989; Minca 2005), the relationship between scientific trials and militarism (Barnes and Farish 2006; Davies 2010; Graham 2011; Kirsch 2007) as well as the prevalence of testing in medicine, education and professional life (Ronell 2005; Rose 1989) all point to troubling overlaps among experiments, power and modern forms of governance, especially when employed to reduce entities and populations to passive targets of experimentation (see Greenhough and Roe 2011; Lemov 2005; Vasudevan 2006). To make sense of this variability of experimentation, we need to engage more closely with a string of experimental sites, starting with a particular space that is most commonly associated with experimentation—the laboratory.

2.1. LABORATORY EXPERIMENTS

Science and technology studies (STS) is an area that explores the mutually shaping relations between culture and nature, science and technology (for overviews, see Law 2008; Powell 2007a; Roberts and Mackenzie 2006). Among the key contributions of this area are detailed accounts of laboratory experiments as well as their cultural, historical and political implications (see Barad 2007; Gieryn 2002; Haraway 1991; Klein 2008; Knorr-Cetina 1992, 1999; Latour 1999; Latour and Woolgar 1986; Pickering 1995; Rheinberger 1997). STS researchers have long questioned the widespread view of laboratories as instrumental and 'placeless' settings, uncontaminated by the outside world and containing standardised equipment (see Kohler 2002). Instead they indicate that laboratories are highly varied spaces, where experimenters are 'tinkering' (Rheinberger 1997, 186) with all kinds of 'apparatuses' (Barad 2007, 168–172) that are 'deliberately arranged to generate surprises' (Gross 2010, 5) in an attempt to bring out new associations and differences in the world. Laboratories involve enacting 'events' (Latour 1999, 126), where scientists, tools, organisms and other nonhuman materials 'perform' (Barad 2007, 49; Pickering 1995, 7) and 'co-produce' (Greenhough 2010, 47) new knowledge about a range of entities and forces whose agency has largely gone unexplored prior to the experiment, whether these, for example, assume the shape of microbial organisms (Latour 1988) or quantum phenomena (Barad 2007). On this view,

the laboratory is not a flat plain upon which life unfolds, but a dynamic space, shaped by and shaping its interaction with the scientists, microbes, machines and other agents who inhabit it (Greenhough 2010, 45).

Laboratory experiments do not so much represent the reality 'out there' as 'make worlds proliferate' (McCormack 2010, 207) by redrawing links between culture, nature and technology. Because the phenomena emerging through experiments are co-produced by the diverse bodies, tools, texts, spaces and nonhuman materials that come together in a specific apparatus, there is a situatedness to experimental knowledge, which 'turns out to be just as local, particular and contingent as anything else' (Roberts and Mackenzie 2006, 158).

According to Hans-Jörg Rheinberger (1997, 80), creating new experimental knowledge is the rare result of a carefully managed process, which involves an ongoing and precarious 'destabilization' of more or less routinised scientific ideas, tools and procedures:

an experimental arrangement must be sufficiently open to generate unprecedented events by incorporating new techniques, instruments, model compounds, and semiotic devices. At the same time it must be sufficiently closed to prevent a breakdown of its reproductive coherence. It has to be kept at the borderline of its breakdown (Rheinberger 1997, 80).

One of the challenges faced by experimenters, then, is how to keep the laboratory arrangement open-ended enough to generate and register new associations and differences in the world. Some arrangements may be too rigid and stabilised to allow for an enactment of unexpected events, which risks turning them 'into devices for testing, into standardized kits, into procedures for making replicas' (Rheinberger 1997, 80). For Latour (2004b, 216), successful experimenters tend to let their set-ups become transformed by the entities they are exploring, whereas poor ones simply force entities to adapt to their preconceptions.

Laboratories necessarily intermingle with their surroundings—for example, they may travel to the 'field', from rainforests to urban spaces, where they usually come to have new functions and parameters, as experiments are adapted to the situated specificities of real-world settings (see Gieryn 2006; Kohler 2002; Latour 1999; Powell 2007b). Likewise, laboratories are imbued with the values and power relations of their social milieu (see Lemov 2005; Vasudevan 2006), while also transforming these by opening new ground for 'ontological politics' (Mol 1999) and 'cosmopolitics' (Stengers 2010) that pose fundamental questions about the kinds of 'collectives' (Latour 1999, 2004a) and 'entanglements' (Barad 2007) humans may form with the more-than-human entities and forces passing through the world. For this reason, laboratory experiments ask for a specific kind of ethical reflexivity from scientists—the skill of 'being responsive to the liveliness of the world' (Juelskær and Schwennesen 2012, 22), which recognises that science and, indeed, the 'human', constitute simply one 'part of the larger material configuration' of the universe (Barad 2007, 171).

2.2. AESTHETIC EXPERIMENTS

Artistic settings do not necessarily seek to replicate the laboratory model of experimentation, but they may 'have as much rigour as any other experimental set-up' (Thrift 2007, 12), employing carefully crafted socio-material arrangements for shaping novel entities, events and techniques of expression, as indicated by recent geographical work on environmental

art (Thornes 2008), literature (Lorimer 2012), animated photography and film (Clarke and Doel 2007), dance (McCormack 2008a) and bioart (Abrahamsson and Abrahamsson 2010) (for overviews, see Hawkins 2011; Last 2012). Scientific sites have also figured in various types of artistic experiments, as is evident from past collaborations between artists and scientists on perceptual phenomena (Smith 2012), the ongoing exchange of influences between science and architecture (Galison and Thompson 1999; Picon and Ponte 2003), and recent aesthetico-theoretical interventions into bioengineering (da Costa and Philip 2008) and military science (Davies 2010).

Aesthetic experiments have their own geographies, requiring 'care, tinkering and labour' (Ingram 2012, 129) as well as specific 'generative constraints' (McCormack 2010, 207) that hold the diverse bodies and materials of the experiment together while also allowing them to interact in surprising ways—for instance, dancers may utilise different conceptions of choreography, from 'open' to 'closed' scores, in order to modulate relations between performers, spaces and audiences (Merriman 2010, 440). Another example of such experimentation is the exhibition space, which can be seen as

a site for the generation rather than reproduction of knowledge and experience. [It is] a kind of laboratory, in which [...] various "actants" (visitors, curators, objects, technologies, institutional and architectural spaces, and so forth) are brought into relation with each other with no sure sense of what the result will be (Basu and Macdonald 2006, 2–3).

Performance art shares this interest in the enactment of unexpected events. Ingram (2012) explores the 'experimental geopolitics' of Wafaa Bilal, who spent one month in a room with a webcam and a paintball gun that could be controlled through a specifically designed webpage. *Domestic tension* may be taken as an 'experimental apparatus' (Ingram 2012, 129), deliberately staged to generate surprising encounters between Bilal and the online participants who were often shooting him with the paintball gun and posting comments in the chat room. Although the experiment could be seen as bringing out the 'inherent violence of a dominant geopolitical culture' (Ingram 2012, 130), it also allowed 'alternative geopolitical practices to emerge' (Ingram 2012, 131)—for example, some participants took over the controls of the paintball gun in an attempt to direct it away from Bilal. For Ingram (2012, 131), this performative experiment 'rendered the small big and the mundane profound, and thereby enacted the possibility of a more pacific geopolitics.'

Aesthetic experiments, then, are often composed to reconfigure everyday practices and spaces by creating new forms of bodily, political and social engagement. Pinder (2005a, 2005b), for instance, has explored artistic interventions into public spaces that are evocative of the experimental urbanism of the Situationist International movement. Such interventions include 'algorithmic walking' tours (Pinder 2005a, 396–397) that use diverse 'navigational devices', such as specific patterns and rules, to generate alternative ways of engaging with urban space, while also 'provoking debate about how [cities] might be different, *better*' (Pinder 2005a, 399; original emphasis; see Hawkins 2011). Community art projects can be understood similarly as 'aesthetic interventions in the affective realm' of daily life, rearranging 'people's embodied movements through and attachments with spaces' as well as widening participation by involving collectives and publics in the improvement and redesign of their environments (Merriman 2010, 440). The content and form of these experimental set-ups may differ from scientific practices, but they are equally important arenas for crafting new geographies, new modes of collectiveness and new ethical sensibilities.

2.3. URBAN EXPERIMENTS

Although the notion of experiment has figured widely in modernist urban imaginaries (see Alison et al. 2007; Gieryn 2006; Heathcott 2005; Kargon and Molella 2008; Pinder 2005b), it has only recently made a more sustained appearance in attempts by geographers to explore cities as emergent and heterogeneous entities, constituted in assemblages of bodies, materials and spaces from near and far, as well as characterised by shifting degrees of openness and stability (see Bender and Farias 2009; Evans and Karvonen 2011; Graham 2011; Gross 2010; Hinchliffe et al. 2005; Latour and Hermant 1998; McFarlane 2011). Cities are increasingly understood as places of ‘cosmopolitical experiments’ (Hinchliffe et al. 2005; see Stengers 2010) and as ‘key sites in testing new ground’ (Amin and Thrift 2002, 156) for collective life, bringing together all kinds of people and objects, animals, organisms and infrastructures, whose encounters have the potential to generate ‘continuous novelty’ (Amin and Thrift 2002, 27).

Early 20th century members of the Chicago School were among the first social researchers to approach urban space as an experimental site (see Gieryn 2006; Gross 2009; Gross and Krohn 2005; Guggenheim 2012). Although some of them evoked the notion of experiment to ‘make their approaches more objective by attempting to adopt the language and methodology of natural sciences’ (Gross and Krohn 2005, 66), not all researchers regarded themselves as scientists in ‘white coats’ (Gross and Krohn 2005, 79) who were conducting experiments *on* the city, but rather as co-experimenters *with* other citizens, experts and communities, who together were seeking new ways of responding to the dynamic qualities of the modern society and its institutional arrangements:

When characterizing the city as an experiment, [researchers] did not merely mean the experiments of city planners or social workers who take society as their object of study, but rather the experimental character of social action and societal development, which—taken full strength—takes the form of an open-ended experiment. (Gross and Krohn 2005, 77)

Illustrative of this thinking is Jane Addams, a Chicago sociologist and activist, who practised ‘experimental social work’ (Gross 2009, 83) at the Hull House settlement for disadvantaged people. For Addams, Hull House was a flexible socio-material arrangement that allowed residents to collaborate with social workers and scientists to improve their living conditions and engage in social transformation through innovative projects, including theatre groups, outdoors schools and co-operative enterprises. The residents were seen as co-experimenters alongside the researchers, who would collectively try out ‘assumptions about social life under realistic conditions’ and then feed findings from these experiments into policymaking, social research and various community initiatives (Gross 2009, 89).

Recent research has added to this work by exploring diverse urban spaces, from green areas to road infrastructures, as ‘living laboratories’ that blur ‘the distinctions between laboratory and field, inside and outside, controlled and uncontrolled’ (Evans and Karvonen 2011, 128; Karvonen and van Heur 2013). Evans and Karvonen (2013), for example, discuss how the Oxford Road corridor in Manchester has been framed by the local government as a space for experimenting with carbon reduction, technological innovation and knowledge-based policy development. Echoing the experiments described in the previous sections, the Oxford Road corridor is here seen as a socio-material set-up, composed of innovative infrastructure and monitoring equipment that collects real-time data about the experiment to be used in policymaking.

Other studies have explored similar ‘climate change experiments’ (Bulkeley and Castán Broto 2013, 362) and ‘green living experiments’ (Marres 2009, 118), whereby experts and laypersons alike are trying out alternative practices of everyday mobility and inhabitation,

including the construction of eco-houses from discarded and renewable materials, which have been studied as a 'seed bed for experimentation' that may help in developing more adaptive and sustainable modes of building in the future (Pickerill and Maxey 2009, 1531). Several public experiments have also been arranged by activists, artists, scientists and local communities, who have sought to involve urban dwellers with their nonhuman environments, for example by raising awareness of the biodiversity of cities through the making of artwork (Hinchliffe and Whatmore 2006) and the collective mapping of animal habitats (Hinchliffe et al. 2005), in this sense employing experimental practices as a way of creating more inclusive urban collectives.

2.4. EXPERIMENTAL DIVERSITY

Taken together, what can the above *geographies of experiment* tell us about actual practices and spaces of experimentation? They suggest that experimental sites are characterised by different conditions and parameters, expectations and outcomes. While one experimental space may certainly become more influential than others, even the most seemingly powerful and standardised site usually carries within itself a rich and sometimes contradictory genealogy of experimentalisms—for instance, Klein (2008) indicates that in the early modern period, a 'laboratory' was a space not only for scientific experimentation but for diverse modes of 'artisanal' innovation, from the distillation of alcohol to the creation of perfumes. This makes it important to explore each site in its empirical specificity and avoid privileging one over the other. The scientific laboratory, or any other single experimental arena for that matter, should therefore not be taken as the archetype of experimentation, but is perhaps best viewed as 'a special case of the general or the real-world experiment in society' (Gross 2009, 93).

At the same time, however, any novel act cannot clearly be taken as experimental—rather, many of the preceding sites appear to share the commitment to compose apparatuses, devices and other *situated* set-ups that induce small variations in the world, all the while observing and recording these variations to learn about their potential (see Karvonen and van Heur 2013). Whether trying out new urban infrastructures, new forms of dance notation or new scientific equipment, experimentation is a careful process of tinkering with relatively limited set-ups of bodies, materials and spaces, which are flexible enough to allow for reconfiguration so as to sustain their transformative potential but also controlled enough to hold together (Rheinberger 1997, 74–83). Although engaging with the unexpected, experimenters are often doing so in a stepwise and recursive manner, trying out different ways of balancing control with openness throughout the process (see Gross 2010; Latour 2004a; Rabinow 2012; Thomke 2003).

Above all, the diverse geographies of experiment point to the challenges involved in evaluating the 'goodness' or 'badness' of an experiment. For example, while inviting more responsive modes of urban governance, sustainability experiments in cities may also reinforce existing social and economic differences, especially when advancing the agendas of those in positions of power and wealth, and when the 'distributions of the risks and benefits' of the experiment are unequal (Bulkeley and Castán Broto 2013, 374; see Evans and Karvonen 2013; Karvonen et al. 2013). Such cases suggest the importance of avoiding crude generalisations about experimentation and accepting that it is only through attending to the *situated* conditions of an experiment as well as the *situated* manner in which its transformations and outcomes are handled that we begin to gain an understanding of its wider political and social implications. This also requires a careful consideration of the kinds of ethical, methodological and theoretical sensibilities that might be conducive to 'good' experimentation—a challenge that will become all the more essential as we now turn to explore how geographers and other researchers have started to rethink their own practices as experimental.

3. *Experimental Geographies*

This part of the article lends insight into work that could be termed *experimental geography*, because it seeks to unfold what the notion of experiment may have to offer for geographical research practices (see Davies 2010; Enigbokan and Patchett 2012; Hawkins 2011; Last 2012). It is interesting to notice that many of the aforementioned studies are not simply describing how others experiment but partake in experiments themselves by tracing out the empirical and ethical, theoretical and methodological possibilities that these open for geographers, encouraging them to 'experiment with the multiple ways in which the world can and is coming to be' (Greenhough 2010, 50).

Apart from having a long and varied methodological career in physical geography (see Richards 2009), experimentation has been debated within the social sciences from the outset (see Dehue 2001; Gieryn 2006; Gross 2009; Guggenheim 2012; Hacking 1999; Howe 2004; Lemov 2005; Manzi 2012). Consistent through these debates is the question to what extent social science may draw on the natural sciences, whose assumed neutrality and objectivity the early social sciences often sought to imitate in order to obtain a sense of 'epistemic authority' (Guggenheim 2012, 100). The 'randomized controlled experiment', where 'particular groups of people are subjected to a treatment and are compared with an untreated control group' (Dehue 2001, 283), was long taken as the ideal type of experimentation, although there are several adaptations of this approach, including field experiments and quasi-experiments, which seek to deal with the specific challenges of running an experiment outside the scientific laboratory (see Dehue 2001; Howe 2004). However, this kind of research has raised concerns about the ethical and methodological viability of experimentation in everyday settings, and has subsequently come to 'represent a mistaken attempt to control, what in essence could not be controlled, namely interactions' (Guggenheim 2012, 108). As a result, many social and cultural researchers have distanced themselves from the notion of experiment, either avoiding it entirely or employing it in a somewhat imprecise metaphorical sense (see Gross 2010; Guggenheim 2012; Powell and Vasudevan 2007).

Nonetheless, we have already seen that there are several experimentalisms at work in everyday environments, and these may often deviate from the ideal of the scientific laboratory and the 'randomised controlled experiment'. The following sections will continue to explore how some of these experimentalisms have shaped more recent methodological and theoretical thinking in human geography and related areas, with a particular emphasis on qualitative and culturally oriented contributions (on quantitative and natural scientific styles of experimentation in the social sciences, see Batty 2007; DeLanda 2011; Manzi 2012; Morgan 2012).

3.1. THEORETICAL EXPERIMENTS

Current performative and nonrepresentational modes of geographical theorising (see Anderson and Harrison 2010; Dewsbury et al. 2002; Latham and Conradson 2003; Thrift 2007) often evoke the notion of experiment to stress the emergent complexities of the world, directing attention to the 'conditions under which something new, as yet unthought, arises' (Rajchman 2000, 17). Instead of making sense of phenomena with prior concepts and ideas, geographers have preferred to approach them as irreducible '*events*', mapping out '*the new potentialities for being, doing and thinking that events may bring forth*' (Anderson and Harrison 2010, 19; original emphasis). This has altered the role of geographical thinking, which no longer strives to simply represent the world as it already appears, but to cultivate new ways of shaping it by 'actively experimenting with the production of space as an integral part of one's own practice' (Paglen 2009, 31).

Many geographers have sought to think through a range of experimental sites, demonstrating the possibilities that these offer for 'enrolling practical activity in the very process of following a train of thought' (Ingold 2011, 15–16). McCormack (2008b, 2) discusses a series of experimental 'thinking-spaces', from performance pieces to architectural compositions, arguing that these can be deliberately arranged to foster 'the processual movement of thought' and to allow this movement to become 'amplified and inflected by novel configurations of ideas, things and bodies'. Greenhough (2010, 50) employs the scientific laboratory to explore geography as a 'performative material intervention' that brings out new ways of engaging with 'more-than-human' entities. Lorimer (2012, 101) describes in a similar fashion the literary experiments of writer Jacquetta Hawkes as 'speculative theory' that offers inspiration for geographical debates on the relations between ecology, materiality and temporality.

These arguments find ample resonance in the influential writings of John Dewey and Gilles Deleuze, which, in their own distinct ways, cultivate the 'art of experimental thinking' (Dewey 2004, 279) to facilitate transformative exchanges between experience and theory. For both of these philosophers, the purpose of thinking is not to occupy a world of its own but to interact with empirical settings in a manner that enables new and more productive associations and differences to emerge. As a result, theoretical labour could be seen as an experimental practice, functioning through specific 'apparatuses' (Barad 2007, 168–172) of concepts, bodies and materials that are devised to 'put the theory at risk' (Latour 2004b, 216) by allowing it to change in ongoing encounters with the world. Developing a similar argument, Amin and Thrift (2005, 223) write that

Theory-making is a hybrid assemblage of testable propositions and probable explanations derived from sensings of the world, the world's persistent ways of talking back, and the effort of abstraction [...] And as such an assemblage, it is always incomplete, always on the mend, and always shot through with inconsistencies. It is a perpetual effort, not a finished template.

As a further example of this kind of geographical thinking, we may take the recent interest in Deleuze-inspired concepts, such as 'affect' (Thrift 2007), 'assemblage' (Anderson and McFarlane 2011) and 'diagram' (McCormack 2005), which have often been mobilised as experimental entities, shaped in dialogue with the world, rather than as fixed theoretical elements with a bounded set of functions and meanings (on concepts, see Bal 2002; Barad 2007; Brown 2002; Deleuze and Guattari 1994; Law 2008; Massumi 2002). Concepts, as Massumi (1992, 24) clarifies their role in the work of Deleuze, 'are logical operators or heuristic devices to be adapted as the situation requires'. Working with concepts is therefore an uncertain 'experimental process which necessarily changes both the field of study and the concept itself' (Malins et al. 2006, 514–515). What makes the above thinking experimental, then, is the demand that it places on geographers and other researchers to continue reassembling and respecifying their conceptual and theoretical set-ups to make them more receptive to the unexplored potential of the world.

3.2. METHODOLOGICAL EXPERIMENTS

There have been plenty of inspiring contributions to the social science methods repertoire of late, including work on creative, digital, mobile, participatory and performative research (see Bates 2011; Davies 2010; Enigbokan and Patchett 2012; Hawkins 2011; Last 2012). This work indicates that experimentation cannot be reduced to one specific approach or technique but invites a more fundamental revision of the very idea of 'method'. Many researchers have begun to take distance from the view of methods as standard equipment that is better suited to one

purpose rather than another. Instead they approach methods as adaptable 'apparatuses' (Barad 2007, 168–172), 'assemblages' (Law 2004, 41–42) or 'devices'—as 'variously constituted, distributed material-semiotic entities' (Lury and Wakeford 2012, 8–10). On such a view, the *a priori* shape assumed by any given method becomes less interesting than the actual transformations that it undergoes during the research process. This turns *all* methods, whether seemingly 'new' or 'old', into sites of potential innovation and researchers and participants into co-experimenters, who are carefully exploring ways of destabilising (Rheinberger 1997, 80; see also Michael 2004) methods to make them more sensitive to the problems at hand. Rather than implying a lack of rigour, this style of working encourages researchers both to recognise the empirical specificity of their methods and to constantly improve them in a recursive and self-correcting dialogue with actual research settings.

We may find several examples of such experimental research across the social sciences, including recent work on the geographies of children and young people, where different set-ups of bodies, materials and spaces have been co-constructed by adults and children to explore ways of widening the expressive and participatory potential of research (see Aarsand and Forsberg 2010; Askins and Pain 2011; Gallacher and Gallagher 2008; Kullman 2012). Krafl and Horton (2007), for example, staged their study as an 'event' by assembling methods so that the participating young people could actively re-shape them through the creative use of different media, from flip charts to cameras and bodily performances. Lenz Taguchi (2010) has worked similarly in educational settings, co-experimenting alongside children and teachers with new pedagogies through the transformative use of mundane materials and substances, such as clay, staircases, sticks and water.

Another aim of this kind of methodological experimentation has been to foster 'collectives' (Latour 2004a) and 'hybrid forums' (Callon et al. 2009) that build on the 'assumption that new capacities, skills, arrangements, and distribution of power may well be required to carry out a successful inquiry' (Rabinow and Bennett 2012, 6). This has engaged researchers in a process of 'redistributing expertise' (Whatmore 2004, 1362), which entails looking for ways to involve activists, artists, scientists and laymen in the co-creation of methods and knowledges relevant to everyday settings, from ecological restoration sites to areas suffering from flooding (see Blue et al. 2012; Davies 2010; Gross 2010; Hinchliffe et al. 2005; Whatmore and Landström 2011). These experiments have not only expanded the participatory potential of geography by inviting researchers to think more inclusively about who and what might take part in knowledge production, they have encouraged

social scientists to ally their research efforts and skills in order to experiment with others (humans and nonhumans) in making new political configurations possible, in bringing new ecological associations and knowledge practices into being (Hinchliffe and Whatmore 2006, 136).

There have also been arguments that experiments could be devised so that all involved parties are allowed to question the socio-material set-up of the enquiry and 'to resist being aligned to only one scientific "truth"' (Whatmore 2003, 98; see also Whatmore 2004, 1362). Being unexpectedly co-shaped by diverse actors and entities, experiments entail, as Stengers (1997; see also Whatmore 2003, 97–98) suggests, 'being at risk', which means that 'what is there will always exceed our capacities to describe, analyse or otherwise engage with it' (Greenhough 2010, 49; see Latour 2004b, 216). Far from constituting a hindrance, the risky aspect of research serves as a reminder of the uncertainties involved in generating new knowledge. As Gross (2010, 5) argues, experimenters need to be willing to cultivate 'a mindful openness to surprises' and accept that experimentation is an uncertain process that constantly 'triggers awareness

of one's own ignorance'. This ignorance, however, may be understood in an affirmative sense as a 'joy of *not* knowing' (Whatmore 2003, 98; original emphasis) that invites ongoing learning.

Indeed, learning comes across as an essential methodological component in recent writings on experimental research across the social sciences—for example, one needs to learn 'to take responsibility for the sometimes surprising side-effects' of experiments (Gross 2010, 169). Likewise, one needs to '*learn to be affected*' (Latour 2004b, 205; original emphasis) by the actors and entities taking part in the experiment, as well as respond flexibly to their shifting characteristics and concerns (see Greenhough and Roe 2011; Hinchliffe et al. 2005). And above all, one has to learn to revise one's thinking by recursively incorporating new knowledge into the research design:

An unsuccessful experiment is not a failed experiment. [...] It might demonstrate that aspects of the experiment needed to be modified so as to *perform* it better. [...] It points to the degree of *skill* or *incapacities* of the researchers, thereby opening up pathways for renewed *training*, *exercise*, and *remediation* of the *practices* involved in the experiment (Rabinow and Bennett 2012, 95; original emphasis).

There is an ethical dimension, then, to experimental learning, and it is the manner in which this learning is handled that opens a potential avenue to distinguish 'good' from 'bad' experimentation (see Latour 2004b; Rabinow 2012; Rabinow and Bennett 2012). Because experimentation fosters emergent entities and events, and is as likely to result in failure as in success, researchers have to exercise constant reflexivity not only by being responsive to the unexpected, but also by making explicit the "learning curve" (Latour 2004a, 194–200) through which they have passed during the research process. Being able to trace back the steps taken along the way and compare the situation *before* and *after* the experiment makes it possible to explore whether something new was created, and if it brought about improvement or not (see Latour 2004b, 219). This also suggests a way for researchers to avoid experimenting simply for the sake of experimentation (see Last 2012, 718), as it implies that experimentation does not solely involve enacting novel events and entities, but exploring how these add to the flourishing of the areas and traditions within which one is working (see Gross 2010; Rabinow and Bennett 2012).

4. Conclusion: A Pedagogy of Experimentation

The previous sections have explored experimentation both as an empirical object of geographical study and as an orientation to geographical research. While experimentation cannot be reduced to one location or arena of practice, most settings discussed in this article appear to share an interest in creating 'sufficiently open' and 'sufficiently closed' (Rheinberger 1997, 80) set-ups from the available bodies, materials and spaces in order to bring about transformations in the world. Whether one is engaging with the *geographies of experiment* (studying experimental practices) or *experimental geographies* (practising experimental research), these areas may usefully learn from one another, even if the two do not always have similar concerns. Experimental geographers, for example, could enhance their awareness of the limits and possibilities of experimentation by drawing on case studies in STS and other fields, whereas geographers of experiment might improve their rapport with empirical sites through the methodological innovations of experimental geographers (see also Last 2012).

Nonetheless, while experimental practices may invite creativity, innovation and new knowledge, experimentation is seldom an end in itself or a guarantee of success. As Latour (2004b, 223) argues, the challenges of experimentation actually indicate that 'good science is rare', not least as experimentation usually entails a deliberate attempt to foster the emergence of something unexpected in the world, which, in turn, asks for a willingness to constantly reconfigure one's modes of working by adapting to unforeseen events as well as learning from other arenas of practice. Above all, experimentation could be taken as a reminder that making novel associations and differences is an uncertain process, where not 'anything goes' but rather 'almost nothing goes', as the ongoingness of the world creates ever-new problems for research and its capacities to engage with these in a receptive and responsible manner (Jensen and Lauritsen 2005, 68–69).

One possible path for future work might therefore be to attend more closely to the question of how to cultivate the experimental ethos (Latham and Conradson 2003, 1904) outlined in this article. What seems particularly valuable in recent writings on experimentation is the space that they offer for a renewed thinking about the role of pedagogy in geographical praxis. Here, pedagogy could be understood as the continuous and lifelong fostering of 'reflective processes by which one becomes capable of flourishing' rather than the mere 'reproduction of knowledge and technique' (Rabinow and Bennett 2012, 44). The work discussed in this article seems to suggest that, if practised with care in an environment conducive to such care, experimentation may well offer an ethically, methodologically and theoretically sensitive way for geography to continue varying alongside the world, while also bringing out its transformative potential.

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Short Biography

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Notes

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¹ After this text was accepted, *Geography Compass* published a thoughtful and highly recommendable article on experimental geography by Angela Last (2012), which also offers detailed insight into the varied challenges and possibilities of this promising area.

² There are plenty of inspiring online resources on experimentation (see also Hawkins 2011; Last 2012), including these websites: <http://merlepatchett.wordpress.com>, <http://www.michaelgallagher.co.uk/>, <http://www.spacesofexperimentation.net/>, <http://mutablematter.wordpress.com/>.

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